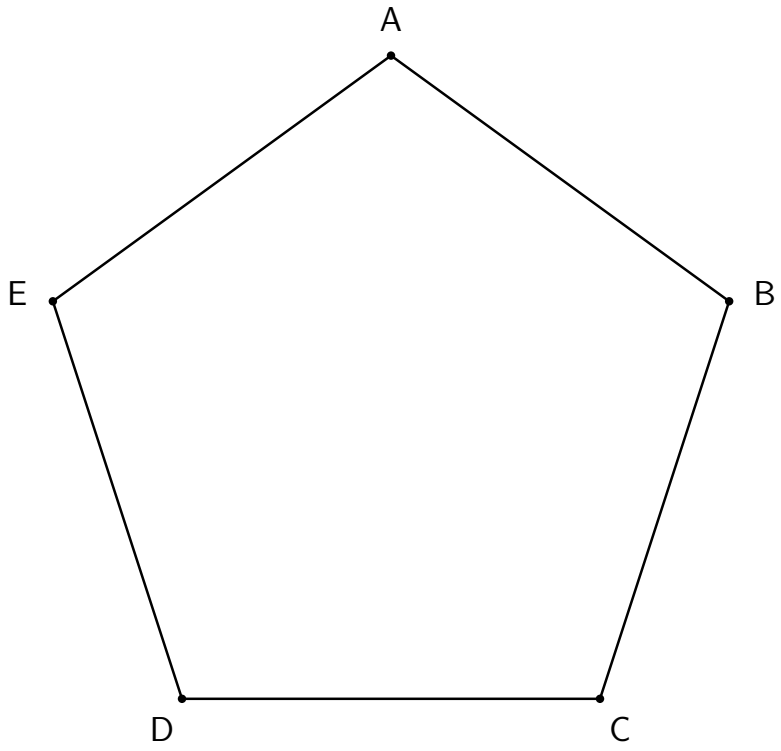
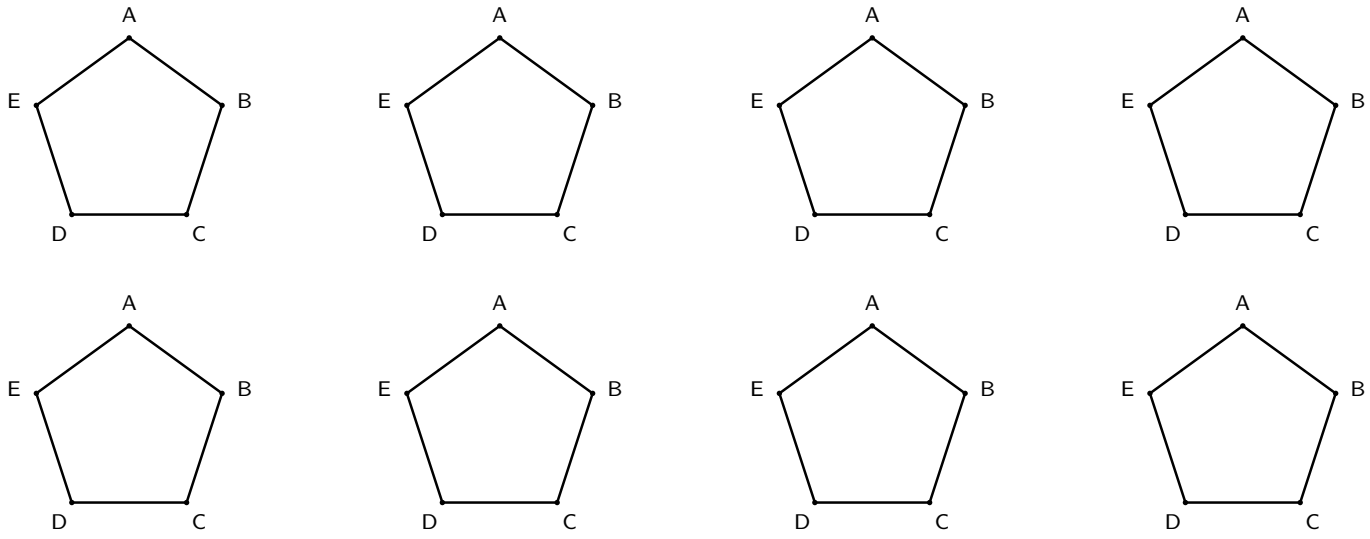
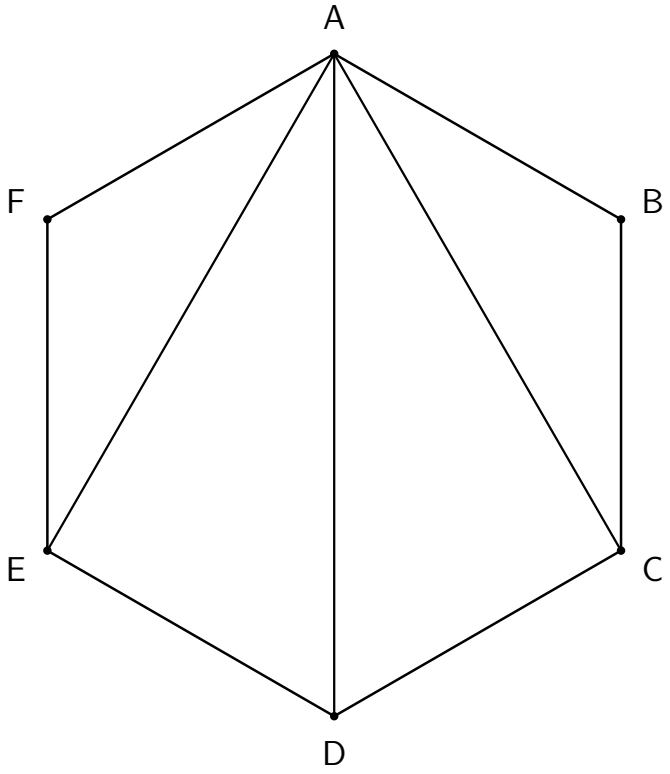
Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

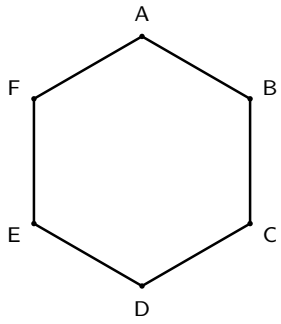
Problem 1: Find every filling of the five-corner polygon. Explain how your collection shows that none are missing.

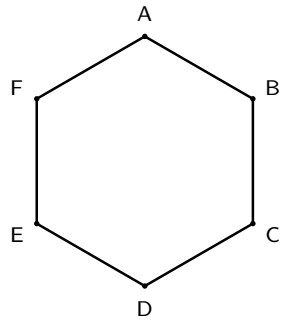


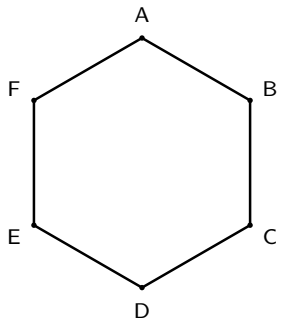


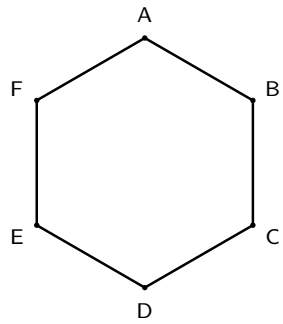
Problem 2: Erase one inside line and replace it with a different line so the polygon stays filled with triangles. Find every result of one such change from each drawing.

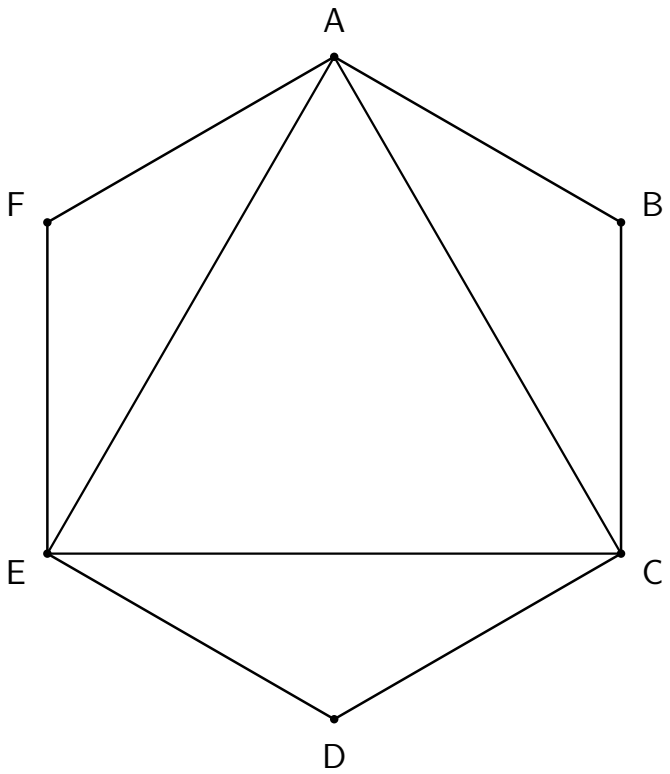


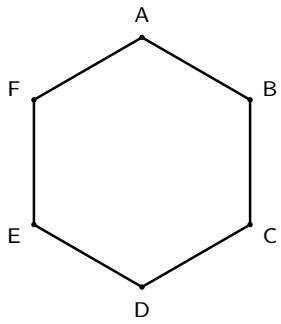


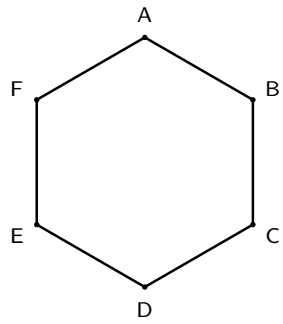


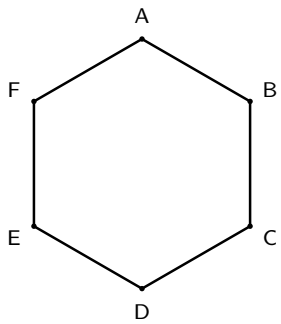


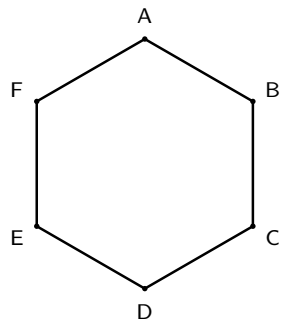




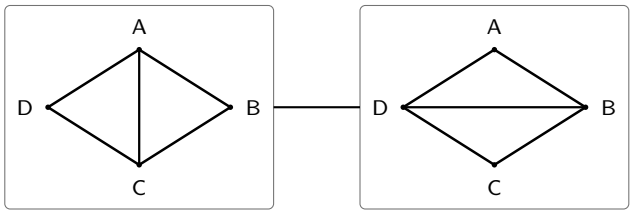








Problem 3: A change of one inside line that keeps the polygon filled with triangles is called a flip. Join two drawings when one flip changes one into the other. Include every possible join. Can you return to a filling after an odd number of flips? Show a route, or explain why it is impossible.



Example: erase AC, draw BD. The join connects whole fillings; it is not an inside line.

